BUS63-4

Strategic Allocation

Committing scarce resources when prediction is cheap and judgment is the binding constraint

The Allocate stage of the Strategy arc.

Albert School — 22 hours

Preface

This course is the Allocate stage of the Strategy arc (Build → Compete → Evolve → Allocate). It assumes the prerequisites and does not re-teach them: ROIC, free cash flow, and DuPont decomposition (BUS23-3); unit economics, defensibility, and the value-creation/value-capture distinction (BUS23-2); NPV, WACC, cost of capital, capital structure, and payout policy (BUS33-3); DCF, scenario analysis, and real-options intuition (BUS43-3); Five Forces, VRIO, generic strategies, and competitive dynamics (BUS33-2). Where this course shares the capital-allocation question with BUS63-1 Corporate Finance Leadership, it takes the general-management posture rather than the finance-function posture.

The conceptual anchor is the Agrawal–Gans–Goldfarb framework: artificial intelligence makes prediction cheap, and human judgment becomes the scarce complementary asset. The terminal deliverable is a strategic allocation audit of a real firm.

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1 ROIC as a strategic allocation compass

1.1 Calculation and interpretation

Return on invested capital measures the operational return a firm earns on the capital deployed in its business, independent of how that capital is financed:

$$\text{ROIC} = \frac{\text{NOPAT}}{\text{Invested capital}} = \frac{\text{EBIT} \times (1 - t)}{\text{Invested capital}}$$

where NOPAT is net operating profit after tax and invested capital is the sum of the capital the firm has put to work (typically equity plus interest-bearing debt, net of non-operating cash). The calculation mechanics are established in BUS23-3; this course uses ROIC as a compass for allocation decisions rather than as a diagnostic endpoint.

1.2 The value-creation test

A firm creates value through growth only when it earns a return above its cost of capital. The test compares ROIC to WACC (established in BUS33-3):

- ROIC $>$ WACC: growth creates value;
- ROIC $=$ WACC: growth is value-neutral;
- ROIC $<$ WACC: growth destroys value.

> **Definition 1** **Value-destroying growth** is revenue or asset growth funded by capital whose return is below the firm's cost of that capital. Reported growth rises while value falls.

1.3 Diagnosis

To diagnose whether reported growth destroys value:

1. Compute ROIC and WACC on the same invested-capital base.
2. Compare them. If ROIC $<$ WACC, additional growth consumes value.
3. Distinguish **accounting growth** (revenue, earnings, asset totals rising) from **returns above the cost of capital**. The two can move in opposite directions: a firm can grow revenue every year while its ROIC sits below WACC, destroying value on every incremental unit of invested capital.

1.4 Limits

ROIC is a backward-looking accounting measure. It is sensitive to the definition of invested capital, to accounting treatments (the comparability adjustments of BUS23-3 — R&D capitalisation, lease treatment, stock-based compensation — apply here), and to timing: capital deployed today may earn returns only in later periods, so a single-period ROIC can understate the return on long-gestation investments.

2 Make, buy, or partner at corporate scale and corporate scope

2.1 The decision

At corporate scale, a firm choosing how to obtain a capability or enter an activity has three modes:

- **make** — build the capability internally (organic investment);
- **buy** — acquire it (covered as M&A in the next chapter);

- **partner** — obtain it through an alliance, joint venture, or contract.

2.2 Beyond cost calculation

The decision is not reducible to which mode is cheapest. It must weigh:

- **strategic coherence** — whether the activity fits the firm's existing activity system and declared strategy (BUS33-2);
- **flexibility** — whether the mode preserves or forecloses future options (developed under real options in Chapter 8);
- **competitive exposure** — whether the mode creates dependence on, or hands advantage to, a rival or a partner who may later become one.

2.3 Corporate scope

Corporate scope is the set of businesses, geographies, and activities the firm chooses to be in. The scope decisions in this course are:

- **business mix** — which businesses the firm operates;
- **geography** — which markets it serves;
- **vertical integration** — which stages of the value chain it owns versus buys (a make/buy decision along the chain);
- **portfolio management** — allocating capital and attention across the business mix;
- **divestiture discipline** — exiting businesses that no longer earn their cost of capital or no longer cohere with strategy.

> **Definition 2** **Divestiture discipline** is the practice of exiting a business when it ceases to create value or to fit the firm's strategy, rather than retaining it by default. Its absence is a common source of value-destroying scope.

3 M&A as a strategic instrument

3.1 When to acquire, build, or partner

Acquisition is one of the three modes from the previous chapter, chosen when the target capability or position cannot be built quickly enough internally (make) and cannot be adequately secured by alliance (partner). The instrument should be selected against the same three criteria — strategic coherence, flexibility, competitive exposure — not on the availability or apparent cheapness of a target.

3.2 The empirical evidence on deal value destruction

The empirical record on acquisitions is that a large share destroy value for the acquirer. The recurring mechanisms are:

- **overpayment** — paying a control premium that exceeds the value of realisable synergies;
- **overestimated synergies** — synergies that are smaller, later, or costlier to realise than assumed;
- **integration failure** — the combined organisation fails to capture the value the deal assumed.

3.3 Diagnosing a capital allocation decision

To diagnose a reported large-firm capital allocation decision — an acquisition, divestiture, buyback, or organic investment — as value-creating or value-destroying:

1. State the decision and the capital committed.
2. Name the **causal mechanism** by which value is created or destroyed (e.g. a control premium exceeding synergies; a buyback executed above intrinsic value; organic investment earning below WACC).
3. State the **counterfactual** — what the firm would have done with the same capital instead, and whether that alternative would have created more value.

4 Executive attention as a scarce resource

4.1 Simon's economics of attention

Herbert Simon's observation is that when information is abundant, the scarce resource is the attention required to process it. Executive attention is therefore a constrained input to be allocated, like capital.

> **Definition 3** **The economics of attention**: a wealth of information creates a poverty of attention. The binding constraint on decision-making in an information-rich organisation is the limited attention of those who must decide.

4.2 Centralisation versus delegation

Because attention is scarce, decisions must be distributed. Centralisation concentrates decisions at the top, conserving coherence but consuming scarce executive attention and creating bottlenecks. Delegation pushes decisions down, conserving executive attention but risking loss of coherence. The allocation question is which decisions to centralise and which to delegate.

4.3 Hierarchy overload

> **Definition 4** **Hierarchy overload** occurs when more decisions are routed to a level of the hierarchy than its attention capacity can process with quality, degrading the quality of every decision at that level.

To analyse a firm's decision-making architecture under attention constraints:

1. Identify where decisions are made and at what volume.
2. Identify where hierarchy overload is degrading decision quality.
3. State what should be centralised (decisions requiring coherence or carrying firm-wide consequence) and what should be delegated (decisions whose local information outweighs the value of central coherence).

5 AI as cheap prediction: the Agrawal–Gans–Goldfarb framework

5.1 Prediction and judgment

The framework of Agrawal, Gans, and Goldfarb separates a decision into two components:

> **Definition 5** **Prediction** is the generation of information about what will happen — filling in missing information from available information. **Judgment** is the determination of the

> relative payoff of different outcomes — what the decision-maker values, and therefore what to do given the prediction.

Artificial intelligence drives down the cost of prediction. When prediction becomes cheap, its complement — judgment — becomes the scarce and valuable input. This is the course's anchor: as AI makes prediction cheap, human judgment becomes the binding scarce asset.

5.2 Classifying decisions

To apply the framework to a real organisation:

1. Decompose each decision into its prediction component and its judgment component.
2. **Delegate to AI** decisions, or the parts of decisions, that are dominated by prediction and whose judgment (the payoffs) is stable and specifiable.
3. **Preserve for human judgment** decisions whose payoffs are contested, novel, or value-laden, where the judgment component dominates.

5.3 Defending the human-judgment boundary

The boundary between what is delegated to AI and what is preserved for human judgment is not only a technical question of where prediction stops. Some decisions should be preserved for human judgment as a **strategic-ethical commitment**, even when they are technically delegable. The defence rests on:

- **competitive externalities** — consequences that fall outside the firm: labour displacement, winner-takes-most dynamics, and innovation-ecosystem effects;
- **accountability consequences** — decisions whose outcomes the board must answer for, and which therefore require a human to hold responsibility.

A decision may be technically delegable to AI yet belong on the human side of the boundary because the board should remain accountable for it.

6 Organisational redesign for the AI era

6.1 Reconfiguring decision structures

If AI shifts where prediction is cheap and judgment is scarce, the decision structures built around the old cost of prediction become misaligned. Redesign reconfigures **which decisions are made where**, and **which are delegated to machines versus preserved for humans** (Chapter 5).

6.2 New sources of managerial leverage

When prediction is cheap, managerial leverage shifts away from the gathering and processing of information and toward the exercise of judgment — specifying payoffs, setting the objectives that predictions serve, and deciding which decisions to preserve. To propose a redesign for a firm deploying AI at scale:

1. Specify which decision structures to reconfigure.
2. State where managerial leverage shifts as a result.

7 Measurable risk versus deep uncertainty

7.1 The distinction

Definition 6 **Measurable risk** is a situation in which the possible outcomes and their probabilities are known or estimable. **Deep uncertainty** (radical ambiguity) is a situation in which the outcomes, their probabilities, or both cannot be specified.

The allocation rule depends on which situation applies:

- under **measurable risk**, **optimisation** is appropriate — maximise expected value given the known probability distribution;
- under **deep uncertainty**, optimisation is unsafe because the distribution it optimises against is not known; **robustness** is appropriate — choose actions that perform acceptably across the range of possible futures rather than optimally against one assumed future.

Common pitfall Optimising against a probability distribution that does not exist — treating deep uncertainty as if it were measurable risk — produces a precise answer to the wrong problem. The first step is to classify the situation, not to compute.

7.2 Robust versus optimised strategies

A robust strategy sacrifices peak performance in the expected scenario for acceptable performance across many scenarios. An optimised strategy maximises performance in the scenario it assumes, at the cost of fragility if that scenario does not occur.

To compare them for a firm facing turbulent conditions, name the scenarios under which each outperforms:

- the optimised strategy outperforms when the assumed scenario materialises;
- the robust strategy outperforms when the future departs from any single assumed scenario — i.e. under genuine turbulence.

8 Real-options reasoning

8.1 Value of flexibility

Real-options reasoning (introduced as intuition in BUS43-3) treats a strategic commitment as analogous to a financial option: the right, but not the obligation, to act later. Flexibility — the ability to delay, expand, contract, or abandon — has value because it lets the firm act on information that arrives after the initial decision.

8.2 Commit versus preserve optionality

Definition 7 The **commit-versus-preserve-optionality threshold** is the point at which the value of committing now (e.g. pre-empting a rival, capturing scale) exceeds the value of waiting to preserve flexibility. Below the threshold, preserve optionality; above it, commit.

8.3 Cost of irreversibility and exit conditions

An irreversible commitment forecloses options: once made, it cannot be costlessly undone. The **cost of irreversibility** is the value of the flexibility given up by committing.

To apply real-options reasoning to an irreversible strategic commitment:

1. Name the **value of flexibility** the commitment would consume.
2. State the **commit-versus-preserve-optionality threshold** — the condition under which committing now beats waiting.
3. Specify **explicit exit conditions** — the observable conditions that would invalidate the commitment and trigger withdrawal, fixed in advance rather than rationalised away later.

9 The strategic allocation audit

The terminal deliverable integrates the course. A strategic allocation audit of a real organisation tests the **coherence** among four things:

1. the firm's **declared strategy** (the competitive logic, from BUS33-2);
2. its **capital allocation** (where capital is actually deployed — judged against ROIC versus WACC, Chapter 1);
3. its **organisational design** (where decisions are made, under attention constraints and AI deployment, Chapters 4–6);
4. its **make/buy/partner decisions** (corporate scope, Chapters 2–3).

The audit is value-creating when these four cohere and value-destroying where they conflict — for example, a declared strategy of focus contradicted by capital allocated to incoherent acquisitions, or a centralised decision structure contradicting a strategy that depends on local responsiveness.

The deliverable must carry:

- **quantified recommendations** — grounded in the financial measures the prerequisites established, not assertions;
- **explicit invalidation conditions** — the observable conditions under which each recommendation would no longer hold (the exit-condition discipline of Chapter 8 applied to the audit itself).

It is produced and defended before a jury.

Through-line. Prediction is cheap; the binding scarce assets are capital, attention, and — above all — human judgment. Strategic allocation is the discipline of committing those scarce assets coherently: deploying capital only above the cost of capital, routing decisions to where attention and judgment are best spent, preserving the decisions a board must answer for, and committing irreversibly only when the value of doing so exceeds the value of the flexibility surrendered.